

A LEAN FORMALIZATION OF SEGAL’S Γ -SETS AS $\mathbb{F}_1$ -MODULES: HYPER-OPERATIONS, THE NERVE ADJUNCTION AND THE SMASH PRODUCT

ABSTRACT. We describe the content of a Lean 4/Mathlib formalization of the elementary theory of Segal’s Γ -sets, regarded (following Connes–Consani) as modules over the “field with one element” $\mathbb{F}_1$. The formalization covers: the Eilenberg–Mac Lane functor and its full faithfulness; the spherical functor and its adjunction with evaluation at 1_+ , together with a counterexample showing that the reducedness condition is necessary; the n -ary hyper-operations on the points of a Γ -set introduced by L. Xu, and a law of generalized associativity which makes these points into what we call an $\mathbb{F}_1$ -module; a nerve construction giving a right adjoint to the points functor; the enveloping commutative monoid of an $\mathbb{F}_1$ -module; and the monoidal structure on Γ -sets given by the smash product. All results are proved without `sorry`, in Lean 4 [9] on top of Mathlib [8].

CONTENTS

1. Introduction	1
Terminology warning	2
Conventions	2
2. The category $\mathcal{N} = \Gamma^{\text{op}}$ and Γ -sets	2
2.1. Pointed finite sets	2
2.2. Pre- Γ -sets and Γ -sets	2
3. The Eilenberg–Mac Lane Γ -set	3
4. The spherical functor	3
4.1. Smash products of pointed sets	3
4.2. The adjunction	3
4.3. Reducedness is necessary	4
5. Hyper-additive structures	4
6. The hyper-operations of a Γ -set	4
7. $\mathbb{F}_1$ -modules and generalized associativity	5
8. The nerve of an $\mathbb{F}_1$ -module	5
9. The enveloping commutative monoid	6
10. The smash product of Γ -sets	6
10.1. Smash product on $\mathcal{N}$	6
10.2. Day convolution	7
11. Correspondence with the files	7
References	8

1. INTRODUCTION

Among the many proposals for a “geometry over the field with one element” (see e.g. [5], and [6] for the related theory of hyperrings), the approach of Connes and Consani [2, 3] takes as basic objects Segal’s Γ -sets [10]: pointed-set valued functors on the category of finite pointed sets. Ordinary commutative monoids (and in particular abelian groups and rings, via their additive structure) embed into Γ -sets through the Eilenberg–Mac Lane construction, and the tautological Γ -set $\mathbb{S}: n_+ \mapsto n_+$ plays the role of $\mathbb{F}_1$: it is the unit for the smash

product of Γ -sets [7] (see also [1] for the homotopical theory). One therefore calls Γ -sets $\mathbb{F}_1$ -modules, and monoids for the smash product $\mathbb{F}_1$ -algebras.

In recent work [11], L. Xu analysed the *additive* structure carried by a Γ -set. For a Γ -set X , its underlying set is $X(1_+)$, and every element $z \in X(n_+)$ has n coordinates in $X(1_+)$ and a total sum in $X(1_+)$. Declaring w to be “a sum” of $v_1, \dots, v_n$ when some z has coordinates v_j and total sum w gives a family of n -ary multivalued operations (hyper-operations). These are not strictly associative, but satisfy a *law of generalized associativity*; Xu uses it to construct an extension of scalars $-\otimes_{\mathbb{F}_1} \mathbb{Z}$ left adjoint to the Eilenberg–Mac Lane functor, and later develops localization and affine schemes over $\mathbb{F}_1$ in this language [12].

The Lean files described here formalize this circle of ideas. A distinctive feature is that we isolate an *axiomatic* notion of $\mathbb{F}_1$ -module at the level of points (Definition 7.1): a set with n -ary hyper-operations, a zero, and a regrouping axiom. We prove that the points of every Γ -set form such an object (Theorem 7.3), that the resulting points functor has a right adjoint given by a nerve construction (Theorem 8.2), and that $\mathbb{F}_1$ -modules in this sense have an enveloping commutative monoid (Theorem 9.2). Composing these adjunctions recovers, at the level of commutative monoids, the shape of Xu’s extension of scalars.

Terminology warning. In [11, 2] an “ $\mathbb{F}_1$ -module” is a Γ -set. In the Lean code, the class `F1module` denotes the axiomatic structure on a set of points of Definition 7.1. We keep this distinction throughout: “ Γ -set” refers to the functor, “ $\mathbb{F}_1$ -module” (or `F1module`) to the hyper-additive structure on points.

Conventions. All Γ -sets live in **Type 0**. Throughout, “pointed set” means a set with a distinguished base point, and **Set**_{*} is the category of pointed sets (`Pointed` in Mathlib).

2. THE CATEGORY $\mathcal{N} = \Gamma^{\text{op}}$ AND Γ -SETS

2.1. Pointed finite sets. For $n \in \mathbb{N}$ let $n_+ = \{0, 1, \dots, n\}$, pointed at 0. The category $\mathcal{N}$ has objects n_+ and morphisms the pointed maps. This is (a skeleton of) the category of finite pointed sets, i.e. Segal’s Γ^{op} . In Lean, n_+ is `Fin (n+1)` pointed at 0.

Lean: `N, N.mk, N.offun` in `N.lean`.

We use the following special maps.

- The *fold* (or sum) map $\sigma_n: n_+ \rightarrow 1_+$, $i \mapsto 1$ for $i \neq 0$ (`N.map_add`).
- The *projections* $p_j: n_+ \rightarrow 1_+$, $j \mapsto 1$ and $i \mapsto 0$ for $i \neq j$ (`N.map_proj`).
- The maps $n_+ \rightarrow 0_+ \rightarrow m_+$ (`N.toZero`, `N.fromZero`); 0_+ is a zero object.
- For $i \in n_+$, the map $\text{pt}_i: 1_+ \rightarrow n_+$, $1 \mapsto i$ (`N.pt`). Note that pt_0 factors through 0_+ .

2.2. Pre- Γ -sets and Γ -sets.

Definition 2.1. A *pre- Γ -set* is a functor $F: \mathcal{N} \rightarrow \mathbf{Set}_*$. It is *reduced* if $F(0_+)$ is a single point. A *Γ -set* is a reduced pre- Γ -set; morphisms are natural transformations. We write **$\Gamma\mathbf{Set}$** for the category of Γ -sets.

Lean: `preGammaSet, preGammaSet.Reduced, GammaSet` in `Gamma.lean`.

The reducedness condition is Segal’s: a Γ -object in a pointed category is a functor preserving the zero object, and 0_+ and $*$ are the zero objects of $\mathcal{N}$ and **$\mathbf{Set}_*$** . Section 4.3 shows that without it basic adjunctions fail.

Example 2.2 (The sphere). The inclusion $\mathbb{S}: \mathcal{N} \rightarrow \mathbf{Set}_*$, $n_+ \mapsto n_+$, is a Γ -set, denoted **$\mathbf{F1}$** in the code. It will be the unit of the smash product (Section 10), and plays the role of $\mathbb{F}_1$.

3. THE EILENBERG–MAC LANE Γ -SET

Definition 3.1. Let M be a commutative monoid. The Γ -set HM has $HM(n_+) = M^n$, pointed at 0, and a pointed map $f: n_+ \rightarrow m_+$ acts by

$$(f_*a)_j = \sum_{\substack{i \geq 1 \\ f(i)=j}} a_i \quad (1 \leq j \leq m).$$

It is reduced since M^0 is a point. An additive map $h: M \rightarrow M'$ acts coordinatewise, giving a functor $H: \mathbf{CMon} \rightarrow \Gamma\mathbf{Set}$.

Lean: `HM, HM.hMap, HMFunctor` in `HM.lean`.

Theorem 3.2. *The functor $H: \mathbf{CMon} \rightarrow \Gamma\mathbf{Set}$ is full and faithful.*

Lean: `HMFunctor.Full, HMFunctor.Faithful` in `HM.lean`.

Sketch. A morphism $\alpha: HM \rightarrow HM'$ has a degree-one component $\alpha_1: M \rightarrow M'$ (`HM.deg1`). Naturality along the collapse maps $n_+ \rightarrow 1_+$ that keep only the j -th coordinate shows that α_n is α_1 applied coordinatewise; naturality along $\sigma_2: 2_+ \rightarrow 1_+$ shows $\alpha_1(x+y) = \alpha_1(x) + \alpha_1(y)$, and naturality at level 0 shows $\alpha_1(0) = 0$. Hence $\alpha = H(\alpha_1)$. $\square$

4. THE SPHERICAL FUNCTOR

4.1. Smash products of pointed sets. For pointed sets $(X, x_0), (Y, y_0)$ the smash product $X \wedge Y$ is $X \times Y$ with the wedge $\{x_0\} \times Y \cup X \times \{y_0\}$ collapsed to a point. Since lying in the wedge is a predicate, the relation “equal or both in the wedge” is already an equivalence relation, and $X \wedge Y$ is constructed as a `Quotient`. The file proves the universal property (maps out of $X \times Y$ constant on the wedge descend) and functoriality.

Lean: `Pointed.smash, Pointed.Smash.lift, Pointed.Smash.map, Pointed.smashFunctor` in `Smash.lean`.

4.2. The adjunction. For a pointed set X let SX be the pre- Γ -set $n_+ \mapsto X \wedge n_+$, with $f: n_+ \rightarrow m_+$ acting by $\text{id}_X \wedge f$. It is reduced since $X \wedge 0_+ = *$, so we obtain a functor $\text{Sph}_0: \mathbf{Set}_* \rightarrow \Gamma\mathbf{Set}$. Let $\text{Ev}_1: \Gamma\mathbf{Set} \rightarrow \mathbf{Set}_*$ be evaluation at 1_+ .

Lean: `GammaSet.S, GammaSet.Sph, GammaSet.S_reduced, GammaSet.Ev` in `spherical.lean, spherical_adjunction.lean`.

Theorem 4.1. $\text{Sph}_0 \dashv \text{Ev}_1$. *That is, for a pointed set X and a Γ -set F , there is a natural bijection*

$$\text{Hom}_{\Gamma\mathbf{Set}}(SX, F) \cong \text{Hom}_{\mathbf{Set}_*}(X, F(1_+)).$$

The unit is $x \mapsto [x, 1] \in X \wedge 1_+$ and the counit at level n is $[y, i] \mapsto F(\text{pt}_i)(y)$.

Lean: `GammaSet.homEquivOfReduced, GammaSet.adj` in `spherical_adjunction.lean`.

Sketch. Given $\varphi: X \rightarrow F(1_+)$, define $SX(n_+) \rightarrow F(n_+)$ by $[x, i] \mapsto F(\text{pt}_i)(\varphi(x))$. This formula is constant on $\{x_0\} \times n_+$ because φ is pointed, and on $X \times \{0\}$ because F is reduced: pt_0 factors through 0_+ . This is the only place reducedness is used. The two round trips are naturality along pt_i . $\square$

4.3. Reducedness is necessary. For a pointed set A let F_A be the constant pre- Γ -set: $F_A(n_+) = A$ and every map acts as the identity. It is reduced only if $A = *$.

Theorem 4.2. *For every pointed set X and every A , the set $\text{Hom}(SX, F_A)$ of natural transformations is a single point. Consequently there is no bijection $\text{Hom}(SX, F) \cong \text{Hom}(X, F(1_+))$ valid for all pre- Γ -sets F , and the spherical functor $\mathbf{Set}_* \rightarrow \text{Fun}(\mathcal{N}, \mathbf{Set}_*)$ is not left adjoint to evaluation at 1_+ .*

Lean: `GammaSet.uniqueHomConstGamma, GammaSet.no_homEquiv, GammaSet.not_adj in ConstGamma.lean.`

Sketch. Let $z: n_+ \rightarrow n_+$ be the zero map. On SX it acts by $[x, i] \mapsto [x, 0] = *$, on F_A as the identity. Naturality of $\alpha: SX \rightarrow F_A$ along z forces α_n to be constant at the base point. But for $X = A = 1_+$, $\text{Hom}(X, F_A(1_+)) = \text{Hom}(1_+, 1_+)$ contains both the identity and the zero map. $\square$

5. HYPER-ADDITIVE STRUCTURES

Definition 5.1. A *hyper-additive structure* on a set M is a family of maps $\boxplus_n: M^n \rightarrow \mathcal{P}(M)$, $n \geq 0$ (the “set of possible values of the sum”). A (weak) *morphism* $f: M \rightarrow M'$ is a map with

$$f(\boxplus_n(x_1, \dots, x_n)) \subseteq \boxplus_n(f(x_1), \dots, f(x_n))$$

for all n and all tuples. These form a category **HAdd**.

Lean: `HyperAdd, HyperAdd.Hom, HyperAddCat in HyperAdd.lean.`

Every (commutative) monoid has the hyper-additive structure $\boxplus_n(x_1, \dots, x_n) = \{x_1 + \dots + x_n\}$.

Proposition 5.2. *The resulting functor $\mathbf{Mon} \rightarrow \mathbf{HAdd}$ (and $\mathbf{CMon} \rightarrow \mathbf{HAdd}$) is full and faithful.*

Lean: `AddMonCat.toHyperAddCat, HyperAdd.Hom.toAddMonoidHom in HyperAdd.lean.`

The point is that an inclusion of singletons is an equality, so a weak morphism between monoids is additive, and the case $n = 0$ gives $f(0) = 0$.

Remark 5.3 (Why weak morphisms). The file `HyperAddStrict.lean` develops the same theory with *strict* morphisms (= instead of $\subseteq$); it is an alternative to `HyperAdd.lean` and defines the same names. The strict notion is too strong: see Remark 6.3.

6. THE HYPER-OPERATIONS OF A Γ -SET

Definition 6.1 ([11, §3.2]). Let X be a Γ -set and write $X_n = X(n_+)$. For $v \in X_1^n$ put

$$\boxplus_n(v) = \{ X(\sigma_n)(z) \mid z \in X_n, X(p_j)(z) = v_j \text{ for } 1 \leq j \leq n \}.$$

Lean: `GammaSet.vect, GammaSet.hadd, GammaSet.HyperAdd in GammaHyperAdd.lean.`

Proposition 6.2. *A morphism of Γ -sets induces a weak morphism on points, and this gives a functor $\Gamma\mathbf{Set} \rightarrow \mathbf{HAdd}$.*

Lean: `GammaSet.hyperAddHom, GammaSet.toHyperAddCat in GammaHyperAdd.lean.`

Remark 6.3. The inclusion cannot be improved to an equality. For the Hurewicz map $\mathbb{S} \rightarrow H\mathbb{N}$, $i \mapsto e_i$, and $v = (1, 1)$: in $\mathbb{S}$ no $z \in 2_+$ has both coordinates equal to 1, so $\boxplus_2(1, 1) = \emptyset$ (“ $1 + 1$ is undefined in $\mathbb{F}_1$ ”), while in $H\mathbb{N}$ one has $\boxplus_2(1, 1) = \{2\}$. (This example is explained in the documentation of the files, not stated as a Lean lemma.)

7. $\mathbb{F}_1$ -MODULES AND GENERALIZED ASSOCIATIVITY

For a partition p of $\{1, \dots, n\}$ into m blocks $B_1, \dots, B_m$ (partition n m ; blocks may be empty), and subsets $A_1, \dots, A_n \subseteq M$, write $\boxplus_n(A_1, \dots, A_n) = \bigcup_{a_i \in A_i} \boxplus_n(a_1, \dots, a_n)$ and $\boxplus_B(A)$ for the hyper-sum over the elements of a block B listed in increasing order (`HyperAdd.set`, `HyperAdd.finset`).

Definition 7.1. An $\mathbb{F}_1$ -module is a hyper-additive structure M together with an element $0 \in M$ such that

- (i) $0 \in \boxplus_k(0, \dots, 0)$ for every $k \geq 0$;
- (ii) 0 is the only empty sum: $\boxplus_0() \subseteq \{0\}$;
- (iii) (*hyper-associativity*) for every partition $p = (B_1, \dots, B_m)$ of $\{1, \dots, n\}$ and all $A_1, \dots, A_n \subseteq M$,

$$\boxplus_n(A_1, \dots, A_n) \subseteq \boxplus_m(\boxplus_{B_1}(A), \dots, \boxplus_{B_m}(A)).$$

A morphism of $\mathbb{F}_1$ -modules is a weak morphism preserving 0 . We write $\mathbb{F}_1\mathbf{Mod}$ for this category.

Lean: `F1module`, `F1module.Hom` in `F1Modules.lean`.

Lean: `F1moduleCat` in `F1moduleCat.lean`.

Condition (iii) says that any value of a sum can be obtained by first summing along the blocks of a partition and then summing the results. For a commutative monoid it is exactly the fact that a finite sum can be regrouped (`AddCommMonoid.F1module`).

Example 7.2 (Subsets). Let $\mathcal{P}(n)$ be the set of subsets of $\{1, \dots, n\}$, with $\boxplus_k(S_1, \dots, S_k) = \{\bigcup S_i\}$ if the S_i are pairwise disjoint and $\emptyset$ otherwise, and zero $\emptyset$. This is an $\mathbb{F}_1$ -module. It is the set of points of the representable Γ -set $\mathcal{N}(n_+, -)$: a pointed map $n_+ \rightarrow 1_+$ is a subset of $\{1, \dots, n\}$, and a family of such maps lifts to level k exactly when the subsets are disjoint.

Lean: `Subsets.instF1module`, `Subsets.hadd_assoc` in `Subsets.lean`.

Theorem 7.3 (Generalized associativity; cf. [11, §3.5]). *For every Γ -set X , the points $X(1_+)$ with the hyper-operations $\boxplus_n$ and zero the base point form an $\mathbb{F}_1$ -module. Hence the functor of the previous section factors through a functor $\Gamma\mathbf{Set} \rightarrow \mathbb{F}_1\mathbf{Mod}$.*

Lean: `GammaSet.hadd_assoc`, `GammaSet.point_mem_hadd`, `GammaSet.eq_point_of_mem_hadd_zero` in `GammaF1Module.lean`.

Sketch. A partition p gives two kinds of maps in $\mathcal{N}$: the *block map* $b_p: n_+ \rightarrow m_+$, sending i to the index of its block, and for each block B the *restriction* $r_B: n_+ \rightarrow |B|_+$, sending the k -th element of B to k and the rest to 0 . One checks three identities in $\mathcal{N}$:

$$p_k \circ r_B = p_{e(k)}, \quad p_j \circ b_p = \sigma_{|B_j|} \circ r_{B_j}, \quad \sigma_m \circ b_p = \sigma_n$$

(`N.restr_comp_proj`, `N.block_comp_proj`, `N.block_comp_add`). If $z \in X_n$ is a lift of $(a_1, \dots, a_n)$ with sum x , then $X(b_p)(z)$ is a lift at level m with sum x , and $X(r_{B_j})(z)$ exhibits its j -th coordinate as a sum of the a_i , $i \in B_j$. Condition (i) holds because the base point of X_k lifts $(0, \dots, 0)$; condition (ii) is where reducedness is used, since a lift of the empty tuple lives in $X_0 = *$. No other hypothesis on X is needed. $\square$

8. THE NERVE OF AN $\mathbb{F}_1$ -MODULE

Definition 8.1. Let M be an $\mathbb{F}_1$ -module. The *nerve* $\mathbf{Nv} M$ is the Γ -set with

$$\mathbf{Nv} M(n_+) = \mathbf{Hom}_{\mathbf{HAdd}}(\mathcal{P}(n), M)$$

(all weak morphisms of hyper-additive structures), pointed at the constant map 0 . A pointed map $\varphi: n_+ \rightarrow m_+$ acts by precomposition with the preimage map $\mathcal{P}(m) \rightarrow \mathcal{P}(n)$, $T \mapsto \{i \geq 1 : \varphi(i) \in T\}$, which preserves disjoint unions.

Lean: `Nerve`, `Nerve.hMap`, `Nerve.nerveFunctor` in `Nerve.lean`.

This is the usual nerve construction relative to the “model” objects $\mathcal{P}(n)$ of Example 7.2. The constant map 0 is a weak morphism precisely by axiom (i).

Theorem 8.2. *The points functor $\Gamma\mathbf{Set} \rightarrow \mathbb{F}_1\mathbf{Mod}$ of Theorem 7.3 is left adjoint to $\mathrm{Nv}: \mathbb{F}_1\mathbf{Mod} \rightarrow \Gamma\mathbf{Set}$:*

$$\mathrm{Hom}_{\mathbb{F}_1\mathbf{Mod}}(X(1_+), M) \cong \mathrm{Hom}_{\Gamma\mathbf{Set}}(X, \mathrm{Nv} M).$$

Lean: `Nerve.homEquiv`, `Nerve.adj` in `Nerve.lean`.

Sketch. Write $\chi_S: n_+ \rightarrow 1_+$ for the indicator of $S \subseteq \{1, \dots, n\}$. Given $f: X(1_+) \rightarrow M$, send $z \in X_n$ to $S \mapsto f(X(\chi_S)(z))$. This is a weak morphism $\mathcal{P}(n) \rightarrow M$: for a disjoint family $S_1, \dots, S_k$ the separation map $n_+ \rightarrow k_+$, $S_r \mapsto r$, satisfies $p_r \circ \mathrm{sep} = \chi_{S_r}$ and $\sigma_k \circ \mathrm{sep} = \chi_{\bigcup S_r}$, so $X(\mathrm{sep})(z)$ is a lift of $(X(\chi_{S_r})(z))_r$ with sum $X(\chi_{\bigcup S_r})(z)$. Conversely, $\alpha: X \rightarrow \mathrm{Nv} M$ gives $x \mapsto \alpha_1(x)(\{1\})$. The round trips follow from $\chi_{\{1\}} = \mathrm{id}_{1_+}$ and from the fact that the preimage of $\{1\}$ under χ_S is S . The unit $X \rightarrow \mathrm{Nv}(X(1_+))$ records all partial sums $z \mapsto (S \mapsto X(\chi_S)(z))$. $\square$

Remark 8.3. It is essential that the levels of $\mathrm{Nv} M$ consist of *all* weak morphisms, and not only the zero-preserving ones; otherwise the argument of Section 4.3 would break the adjunction for non-reduced objects. For a commutative monoid M , a weak morphism $a: \mathcal{P}(n) \rightarrow M$ is determined by $a(S) = \sum_{i \in S} a(\{i\})$, so $\mathrm{Nv} M \cong HM$; this identification is noted in the documentation but not yet formalized.

9. THE ENVELOPING COMMUTATIVE MONOID

Definition 9.1. Let M be an $\mathbb{F}_1$ -module. Its *enveloping monoid* $\mathrm{Env} M$ is the free commutative monoid on M (finite multisets) modulo the congruence generated by $[c] \sim [x_1] + \dots + [x_n]$ whenever $c \in \boxplus_n(x_1, \dots, x_n)$.

Lean: `F1module.envCon`, `F1module.Env`, `F1module.unit`, `F1module.lift` in `F1moduleAddCommMonCat.lean`.

Theorem 9.2. $\mathrm{Env}: \mathbb{F}_1\mathbf{Mod} \rightarrow \mathbf{CMon}$ is left adjoint to the functor $\mathbf{CMon} \rightarrow \mathbb{F}_1\mathbf{Mod}$ sending a commutative monoid to itself with $\boxplus_n(x) = \{\sum x_i\}$.

Lean: `F1moduleCat.env`, `F1moduleCat.envAdj` in `F1moduleAddCommMonCat.lean`.

Corollary 9.3 (shape). *Composing Theorems 8.2 and 9.2, $\mathrm{Env} \circ \mathrm{Points}: \Gamma\mathbf{Set} \rightarrow \mathbf{CMon}$ is left adjoint to $\mathrm{Nv} \circ \mathrm{incl}: \mathbf{CMon} \rightarrow \Gamma\mathbf{Set}$, which is expected to be isomorphic to H . This is the commutative-monoid analogue of Xu’s extension of scalars $-\otimes_{\mathbb{F}_1} \mathbb{Z} \dashv H$ [11, §4]; composing further with group completion would give the version with abelian groups.*

The composite adjunction is a formal consequence of the two formalized ones, but it is not stated as a single Lean declaration.

10. THE SMASH PRODUCT OF Γ -SETS

10.1. Smash product on $\mathcal{N}$. The smash product $m_+ \wedge n_+$ is realised concretely as $(mn)_+$, via the encoding $\mathrm{enc}: m_+ \times n_+ \rightarrow (mn)_+$ which sends the wedge to 0 and is the standard bijection $\{1..m\} \times \{1..n\} \cong \{1..mn\}$ off it. Maps out of $(mn)_+$ are determined by their values on encoded pairs; this gives $f \wedge g$, associators, unitors and the symmetry of $\mathcal{N}$.

Lean: `N.enc`, `N.dec`, `N.smashMap`, `N.assocN`, `N.lamN`, `N.rhoN`, `N.swapN` in `NMonoidal.lean`.

10.2. Day convolution. Following Day [4] and Lydakis [7], the smash product of Γ -sets is

$$(F \wedge G)(k_+) = \operatorname{colim}_{m_+ \wedge n_+ \rightarrow k_+} F(m_+) \wedge G(n_+).$$

In Lean the colimit is the quotient of the set of quintuples $(m, n, g: m_+ \wedge n_+ \rightarrow k_+, x \in F(m_+), y \in G(n_+))$ by the relation $(m, n, (a \wedge b) \circ g, x, y) \sim (m', n', g, F(a)x, G(b)y)$. Using $F(m_+) \times G(n_+)$ instead of the smash is harmless: for reduced F, G the wedge is automatically collapsed (`Day.mk_point_left`, `Day.mk_point_right`). Morphisms out of $F \wedge G$ correspond to *bimorphisms* $F(m_+) \times G(n_+) \rightarrow H(m_+ \wedge n_+)$ natural in both variables.

Lean: `GammaSet.Day.tensorF`, `GammaSet.Day.Bimor`, `GammaSet.Day.desc` in `GammaMonoidal.lean`.

Theorem 10.1. $(\Gamma\mathbf{Set}, \wedge, \mathbb{S})$ is a monoidal category.

Lean: `GammaSet.Day.associator`, `GammaSet.Day.leftUnitor`, `GammaSet.Day.pentagon`, `instance MonoidalCategory GammaSet` in `GammaMonoidal.lean`.

Method. A morphism out of an iterated smash product is determined by its values on generators $\iota(\iota(x, y), z)$ (`Gens.ext`); on generators every structure map acts through a map of $\mathcal{N}$, and equalities of maps of $\mathcal{N}$ are checked on encoded elements (`N.hom_ext_enc`). The unitors use that $\mathbb{S}$ is the free Γ -set on $1 \in \mathbb{S}(1_+)$. $\square$

Remark 10.2. The symmetric structure and the internal hom $\underline{\operatorname{Hom}}(M, N)(n_+) = \operatorname{Hom}_{\Gamma\mathbf{Set}}(M, N(n_+ \wedge -))$, announced in the header of `GammaMonoidal.lean`, are not yet formalized. Together with them, $\mathbb{F}_1$ -algebras (monoids for $\wedge$) and the multiplicative version of the extension of scalars [11, §5], as well as localization [12], are natural next steps.

11. CORRESPONDENCE WITH THE FILES

File	Content
<code>N.lean</code>	The category $\mathcal{N}$; special maps.
<code>Smash.lean</code>	Smash product of pointed types.
<code>Gamma.lean</code>	Pre- Γ -sets, reducedness, $\Gamma\mathbf{Set}$, the sphere $\mathbb{S}$.
<code>NMonoidal.lean</code>	Smash product on $\mathcal{N}$.
<code>HM.lean</code>	HM ; Theorem 3.2.
<code>spherical.lean</code>	The spherical functor.
<code>spherical_adjunction.lean</code>	Theorem 4.1.
<code>ConstGamma.lean</code>	Counterexample of Section 4.3.
<code>HyperAdd.lean</code>	Hyper-additive structures, weak morphisms.
<code>HyperAddStrict.lean</code>	Variant with strict morphisms.
<code>F1Modules.lean</code>	Definition 7.1.
<code>F1moduleCat.lean</code>	The bundled category $\mathbb{F}_1\mathbf{Mod}$ and the associated functors.
<code>GammaHyperAdd.lean</code>	Hyper-operations of a Γ -set.
<code>GammaF1Module.lean</code>	Theorem 7.3.
<code>Subsets.lean</code>	Example 7.2.
<code>Nerve.lean</code>	Theorem 8.2.
<code>F1moduleAddCommMonCat.lean</code>	Theorem 9.2.
<code>GammaMonoidal.lean</code>	Monoidal structure on $\Gamma\mathbf{Set}$.

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